

EXECUTIVE BRIEF

Sets, Logic, Relations and Functions

From formal definitions to transparent data systems and defensible decisions

Production unit	PU-B01-C02 SRAI Book 1, Lesson 2
Controlled release	v1.0.1 Published 26 August 2026
Author	Mbaye Kebe Creator and Author of SRAI
Audience	Executives, statisticians, analysts, data professionals, system designers and AI/ML practitioners

EXECUTIVE TAKEAWAY

Reliable data systems begin by declaring what belongs, what is true, what is connected and how inputs map to outputs. These are not abstract preliminaries: they are the control layer behind filters, database joins, eligibility rules, knowledge graphs, software functions and explainable automated decisions.

Why this lesson matters

Organisations often treat classifications, logical conditions and mapping rules as implementation details. In practice, weak definitions at this level create silent errors: records enter the wrong population, missing values are treated as false, joins create unsupported relationships, and automated decisions cannot be reconstructed or defended.

Lesson 2 provides a common formal language for statisticians, data engineers, developers, researchers and decision-makers. It turns assumptions into inspectable objects that can be reviewed before they are embedded in code or institutional processes.

The four foundations and their operational meaning

Foundation	Question answered	Operational consequence
Sets	What belongs to the declared universe or category?	Controls population definitions, inclusion/exclusion rules, cohorts, filters and data coverage.
Logic	Under which stated conditions is a proposition true?	Controls validation rules, eligibility decisions, alerts and compound conditions; prevents implication from being mistaken for causation.
Relations	Which entities are connected, and under what rule?	Controls database joins, networks, knowledge graphs, matching systems and traceable associations.
Functions	How does each declared input map to an output?	Controls transformations, scoring rules, software interfaces and whether missing or exceptional inputs are handled explicitly.

Decision-system application

Consider an eligibility system for a public programme. The universe must first be declared: all applicants received before the closing date, or only applicants with complete files? Predicates then express the rules, such as residence, age, income threshold and documentation status. A relation connects applicants to evidence and programme categories. A function or decision rule maps each valid input record to an outcome and a reason code.

When these components are explicit, a reviewer can determine why a case was included, excluded, approved or rejected. When they remain implicit, the same system may produce technically executable but institutionally indefensible results.

Controls leaders should require

- Declare the universe, unit of analysis and admissible values before building filters or models.
- Separate missing, unknown and not-applicable states from false; do not collapse them silently.
- Document the domain, codomain and exceptional cases of every operational mapping.
- Test totality and uniqueness when a process claims to be a function; identify partial mappings deliberately.
- Attach reason codes and retained evidence to automated decisions so outcomes can be reconstructed.
- Version definitions and rules together with code, data and published outputs.

Implications for AI and machine learning

Machine-learning systems inherit the universes, labels, relations and transformations defined upstream. A model can be statistically sophisticated while still operating on an incoherent population, ambiguous labels or undocumented mappings. Formal clarity therefore supports responsible AI by strengthening traceability, reproducibility and contestability before model evaluation begins.

What the controlled release provides

- An audited Chapter 2 publication defining sets, logic, relations, partial functions and total functions.
- A canonical executable notebook with environment reporting, assertions and reference implementations.
- Sixteen graded exercises with complete worked solutions and Python reference implementations.
- Local and clean Google Colab validation evidence, plus owner substantive approval.
- A coordinated long-form YouTube lesson and a versioned GitHub release package.

Release note: v1.0.1

Version 1.0.1 adds the standalone Exercises and Solutions package in DOCX and PDF formats. The approved chapter, notebook, code and lesson content are unchanged from v1.0.0. The release ZIP is protected by SHA-256 digest 3b14a71795ab21d3914fa18fbc7cdb45e5a409a4aef4864b97b909549327b710.

Recommended executive use

Use this Production Unit as a governance and design checkpoint whenever an initiative depends on classification, validation, joining, scoring or automated decisions. Before approving implementation, ask the project team to show the declared universe, predicates, relation rules, input-output mappings, exception handling and decision evidence.

Use case	Executive question	Evidence expected
Survey or administrative data	Who and what are inside the analytical universe?	Population definition, inclusion rules, missing-data treatment.
Database integration	What justifies each link between records?	Keys, cardinality, join rules, unmatched-record controls.
Eligibility or scoring	Can every outcome be explained and reproduced?	Predicates, mapping rule, exceptions, reason codes, audit trail.
AI/ML pipeline	Are labels and transformations formally coherent?	Label definitions, domains, feature mappings, validation assertions.

Controlled resources

Lesson page: <https://srai.mbayekebe.net/learn/pu-b01-c02/>

YouTube lesson: <https://youtu.be/VYXyB1-wUFs>

GitHub release: <https://github.com/mbayekebe/srai-book-01-mathematical-foundations/releases/tag/pu-b01-c02-v1.0.1>

Release package: PU-B01-C02_GitHub_Repository_v1.0.1.zip

Canonical notebook: M1_N02_sets_logic_relations_functions.ipynb

Sources and control basis

- PU-B01-C02 README, controlled release v1.0.1.
- PU-B01-C02 Controlled Release Record, approved final release.
- PU-B01-C02 Validation Report, local and clean Google Colab execution evidence.
- SRAI Lesson 2 website publication and coordinated YouTube lesson.

SRAI - From statistical reasoning to responsible artificial intelligence.